

I. Relativistic Kinematic

1. A spaceship with a rest length of 100 m passes in front of an Earth observer in $4\mu\text{s}$. What is its velocity relative to the Earth?
2. We study an electron in a radioactive particle beam as it passes through the laboratory. It is found that, as measured in the laboratory, the average lifetime of the particle is 20 ns; after this time the particle decays. The average lifetime measured in the reference frame attached to the particles is 7.5 ns . What is the propagation velocity of the particle beam?
3. A certain bacterium doubles its population in 20 days. Two such bacteria are placed in a spaceship and sent on a 1000-day space journey. During the trip, the spaceship moves at a constant speed of $0.995c$. How many bacteria will be on board at the moment of return?
4. Aladár drives a nail into point $P_1(4000\text{ m}, -100\text{ m}, 200\text{ m})$ in the coordinate system K . At that moment, the clock in K shows $t = 15\mu\text{s}$. At the same time in K , Balázs drives another nail into point $P_2(3000\text{ m}, 500\text{ m}, 150\text{ m})$ of the same coordinate system. (The two events occur simultaneously in K .) Calculate the time interval between the two events in a reference frame K' moving at a uniform velocity $v = 2 \cdot 10^8\text{ m/s}$, parallel to the x-axis.

II. Relativistic Dynamics

1. A particle moves with a velocity such that $\frac{v}{c} = 0.99$. Determine the value of $\frac{m}{m_0}$!
2. Determine the relativistic mass and the velocity of an electron whose kinetic energy is 100 keV ($1.6 \cdot 10^{-14}\text{ J}$). The rest mass of the electron is $9.11 \cdot 10^{-31}\text{ kg}$.
3. Calculate the rest energy of an electron, corresponding to its rest mass of $9.11 \cdot 10^{-31}\text{ kg}$, in joules and in electronvolts! ($1\text{ MeV} = 1.6 \cdot 10^{-13}\text{ J}$)

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